



High Reynolds number turbulent flow past a rotating cylinder

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ABSTRACT

High Reynolds-number flow over a rotating cylinder is investigated by two-dimensional numerical computations. The Reynolds-Averaged Navier–Stokes (RANS) equations are solved via the finite-volume method and they are closed by a modified k – ε turbulence model. The spin ratio a is defined as the ratio between the cylinder's circumferential speed velocity to the free-stream varies from 2 to 8. The flow is examined at Reynolds numbers from 5×10^5 to 5×10^6 , which is considered to be an interesting range for industrial flows. Available experimental and numerical data were used to verify the validity of the implemented procedure.

The results revealed stabilization of the acting forces at high spin rates, thus indicating a flowfield with suppressed vortex-shedding activity, as it is expected, in accordance with theoretical considerations in previous studies. Load coefficients were found to be inversely proportional to the Reynolds number for most of the examined rotational rates.

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1. Introduction

Investigations of incompressible flow past a rotating circular cylinder have been conducted by many researchers, based on theoretical, experimental and numerical approaches. Spinning cylinder flows are of great importance in aerodynamics and in design of engineering structures. Rotating cylinders are also devices well recognized for the control of boundary layer flows. There exist many research articles dedicated to the applications of rotating cylinders or to the implementation of other control techniques, such as blowing, suction, surface roughness and others. Some relevant articles are those of Bearman [1], Berger and Willie [2], Gad el Hak and Bushnell [3], Griffin and Hall [4], Sumer and Fredsoe [5] and Zdravkovich [6].

Many studies are focused on the steadily rotating and translating circular cylinder. The first attempts aimed mainly at investigating the well-known Magnus effect. Most of the examined cases found in literature revealed a periodic flowfield, accompanied by vortex-shedding activity. However, a careful literature review reveals that the instability of the flow is closely related to a critical value of the dimensionless ratio $a = \frac{U_\theta}{U_\infty}$, almost independently of Re , where U_θ is the circumferential velocity of the cylinder and U_∞ the free-stream velocity. An early experimental study on flow past a spinning circular cylinder was performed by Prandtl [37]. According to the results and the well known photographs from his work, for low rotational speeds, flow separation downstream of the cylinder is observed. Prandtl concluded that the maximum lift force generated by a spinning cylinder exposed to uniform flow could not exceed the value of 4π . However, the reported results of Tokumaru and Dimotakis [7], among others, contradicted the above statement. They demonstrated that the limit of 4π could be exceeded, thus their computations at $Re = 3.8 \times 10^3$ and $\alpha = 10$ revealed 20% larger lift coefficient than the limit of 4π .

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Nomenclature

$C_{1\varepsilon}, C_{2\varepsilon}$	empirical constants in the turbulence model
C_d	drag coefficient
C_l	lift coefficient
C_p	pressure coefficient
D	diameter of cylinder
G	turbulent flux transport term
G_k	turbulence production
k	turbulence kinetic energy (m^2/s^2)
l_ε	length scale in the viscous-wall region
MUSCL	Monotone Upstream-Centered Scheme for Conservation Laws
N	vector normal to surface
p	mean static pressure (N/m^2)
P	production term
p^∞	mean free-stream pressure (N/m^2)
r	radial position vector
R	Reynolds-stress tensor
Re	Reynolds number
$\tilde{t}$	total-stress tensor
t	time (s)
U, V	velocity components in x and y directions (m/s)
U_∞	free-stream velocity (m/s)
U_i, U_j	mean-velocity components (m/s)
u_i, u_j	velocity-fluctuation components (m/s)
U_r	radial velocity
U_θ	circumferential velocity of the cylinder (m/s)
U_v	turbulence shear stress
x, y	coordinates in horizontal and vertical directions (m)
$\underline{x}$	x position vector
x_i, x_j	orthogonal space coordinates (m)
$\underline{y}$	y position vector
y^+	dimensionless distance

Greek symbols

δ_{ij}	Kronecker delta
ε	turbulence dissipation rate (m^2/s^2)
θ	azimuth angle
μ	laminar molecular viscosity ($\text{kg}/\text{m s}$)
μ_t	turbulence viscosity ($\text{kg}/\text{m s}$)
μ_{tv}	turbulence viscosity in the near-wall region ($\text{kg}/\text{m s}$)
μ_{th}	turbulence viscosity in the outer region ($\text{kg}/\text{m s}$)
ν	kinematic viscosity ($\text{kg}/(\text{m}^2 \text{s})$)
ν_t	turbulence kinematic viscosity ($\text{kg}/(\text{m}^2 \text{s})$)
ρ	density (kg/m^3)
ζ_k	turbulence Prandtl number for kinetic energy of turbulence
ζ_ε	turbulence Prandtl number for dissipation rate of turbulence

Subscripts

d	pertains to drag
i, j	i, j directions
l	pertains to lift
T	turbulent

Low Re -number spinning cylinder flows have been the subject of many studies. A well-organized numerical study was published by Ingham and Tang [8]. In that paper, the Navier–Stokes equations were solved via the finite difference method, in order to examine the asymmetrical flow in uniform viscous liquid, at Re numbers 5 and 20 and dimensionless ratio α from 0 to 0.5. Moreover, steady and unsteady flow at low Reynolds number was investigated numerically by Badr et al. [9]. Similar to the above methodology of Ingham [10], is the study of Ingham and Tang [8], who introduced a new numerical technique coupled with series expansion solutions, to avoid difficulties in satisfying the boundary conditions far away from the cylin-

der. Numerical investigations were performed at Re numbers 5, 20 and $0 \leq \alpha \leq 3$. Ingham and Tang [8] also carried out numerical simulations for two-dimensional steady flow, extending their previous research work to $Re = 60$ and 100 for $0 \leq \alpha \leq 1$. Another numerical attempt at $Re = 200$ was reported by Chen et al. [11]. An explicit pseudo-spectral technique was adopted for resolving the fundamental equations. According to their results, when $\alpha = 3.25$ more than one vortex is shed downstream. Kang [12] also has contributed significant work to this research area. Sequential numerical simulations at Re equal to 40, 60, 100 and 160 in the range of $0 \leq \alpha \leq 2.5$ were performed. It was observed that at $60 \leq Re \leq 160$ the maximum value of a , which favors flow instability, varies logarithmically when plotted against Re . Mittal and Kumar [33], following the work of Chen and Patel [14], performed numerical simulations for $0 \leq \alpha \leq 5$ and $Re = 200$. The incompressible Navier–Stokes equations were solved via the finite-element method. A stability analysis was also conducted by new methods, which led to very important theoretical conclusions, related to the uniqueness of the solution in the aforementioned spectrum of α . However, the most important outcome of that work, was the identification of a new range of α , where the flow returns to an unstable mode. When $a < 1.91$ the flow remains time-dependent, then it becomes steady at higher rotational speeds; but at $4.34 < \alpha < 4.70$ it becomes again unsteady. Other studies are also found in the literature, in which different aspects of spinning cylinder laminar flows are examined. Kang [12], examined the two-dimensional laminar flow past a rotating cylinder with uniform shear. Numerical simulations were performed at $Re = 100$ and dimensionless ratio α up to 5.5. Furthermore, Wang and Tan [15] examined cylinder flows in close proximity to walls by experimental means. The vortex shedding evolution, the distribution of mean-fluctuating velocity correlations and the impact to the flow of the gap height between the cylinder and the wall, were the main tasks discussed.

Moderate- Re cylinder flows ($Re > 1000$), are more difficult to solve for, because three dimensional and turbulence effects become prominent. Badr et al. [16] examined the influence of the cylinder's rotational motion to the time evolution of the flow. Experiments and numerical were performed for $10^3 \leq Re \leq 10^4$ and $0.5 \leq \alpha \leq 3$. Their results showed good agreement between experiments and simulations, except for $\alpha = 3$, where some discrepancies were observed. Another numerical attempt at $Re = 1000$ was demonstrated by Chew et al. [17] using a hybrid vortex method.

The examined dimensionless ratios were between 0 and 6. They concluded, consistently with other research articles, that when α exceeds a critical value close to 2.0, then the vortex shedding vanishes. Nair et al. [13] solved unsteady, two dimensional flow equations applying high order discretization schemes. More specifically, numerical runs were performed at $Re = 200$ with $\alpha = 0.5$ and $\alpha = 1$, $Re = 1000$ with $\alpha = 0.5$ and $Re = 3800$ with $\alpha = 0.5$. Their numerical results were compared with available experimental data from literature and they examined the influence of the discretization schemes on the accuracy of the solution, always as a function of α . Also interesting is the work of Chou [19], since the Navier–Stokes equations were solved in a ‘non-primitive variable form’ at $Re = 10^3$, 10^4 and α up to 3.0, using the vorticity form of the equations, which however is not favored in the literature, due, among others, to the complexity in applying boundary conditions.

High Reynolds number flows over cylinders are complicated because of the coupled action of the shear layer instability and the early development of the fully turbulent attached boundary layer. Also, three dimensional effects are more pronounced compared to laminar flow regimes [20]. Critical Reynolds number for the flow past a smooth cylinder is approximately 3×10^5 . At high sub-critical values, two non-linear processes appear. Close to $Re = 2 \times 10^5$ the flow becomes asymmetric, producing a force in the cross-stream direction. Schewe [21] conducted experiments and proved that mean lift is produced at Re from 2.8 to 3.5×10^5 . Also in this range of Re the drag coefficient drops rapidly. This phenomenon is called drag crisis and it is closely related to the asymmetry of the flow. Drag crisis occurs due to the upstream position of the shear-layer instability point relatively to the transitional-unseparated flow region. The flow reattaches because the shear-layer eddies cause mixing of the boundary layer with the outer flow, along the cylinder's surface. The pressure drag coefficient decreases and the friction terms increase. However, after 3.8×10^5 the drag coefficient increases again due to the action of the turbulence shear stresses.

From the open literature it appears that high Reynolds number flows have been investigated mainly for non-rotating cylinders. Breuer [22] has conducted a comprehensive numerical study for a high subcritical Reynolds number $Re = 1.4 \times 10^5$. Turbulence effects have been accounted for by the Large Eddy Simulation (LES) approach. Multiple grids with different spanwise lengths and various resolutions have been used. Two sub-grid scale models (SGS) were implemented and the equations were solved free of SGS modeling. Mean force coefficients and other mean fluctuating terms have been computed in specific regions of the flow domain. The impacts of the aforementioned grid parameters and models were analyzed extensively. Satisfying agreement was found between measurements and numerical simulations. Catalano et al. [23], carried out LES numerical simulations at Reynolds numbers 5×10^5 , 10^6 and 2×10^6 . Their results were compared with Unsteady Reynolds-Averaged Navier–Stokes (URANS) numerical data, RANS data and available experimental measurements.

The open literature is poor in rotating cylinder flows at high Re regimes. To the authors' best knowledge, there is only one published article, concerning the study of the flow past a spinning cylinder at $Re = 1.4 \times 10^5$ [24]. Karabelas [24] conducted LES simulations at spinning ratios from 0 to 2 and at $Re = 1.4 \times 10^5$. Flow stability is achieved at spinning ratios greater than 1.3. Drag considerably decreases, as expected at this Re number. On the contrary, lift coefficient reaches the value of 5 at $a = 2$.

The present work is an attempt to resolve some aspects of the flow over a rotating cylinder at high rotational rates, not investigated as yet by numerical methods. Turbulence effects and high rotational rates for supercritical Reynolds numbers are the novel issues, that must be resolved, because of their industrial applications and theoretical importance. In this study, supercritical flow past a cylinder is examined, assuming fully turbulent boundary layer from its first attachment to the

cylinder. Three Re numbers are examined, $Re = 5 \times 10^5$, $Re = 10^6$ and $Re = 5 \times 10^6$ and 12 rotational rates from $a = 0$ to $a = 8$. This study extends the investigation of the rotational effects to higher rotational rates than those considered to date.

One of the main objectives of this study is to investigate the effects of the cylinder's rotation on the loads exerted by the fluid to the cylinder. The mean turbulence field and the flow structures are also analyzed as function of Re and a and are compared to laminar patterns. The first step is the validation of the computational code and the investigation of possible dependencies of the mean flow on time. Flow structures are discussed and comparisons are made between turbulent and laminar patterns for a variety of Reynolds numbers and rotational rates a . The computation of the aerodynamic loads and the investigation of the vortical activity are also main tasks of discussion. The contribution of the pressure and viscous components of the forces are evaluated for a range of Re and a , as before. Turbulence analysis follows, referring to the computation and discussion of the Reynolds stress tensor. Another interesting topic addressed and analyzed is the turbulent kinetic energy budget for several azimuth positions and rotational rates. Finally, some conclusions summarize the outcome of all presented work.

2. Numerical modeling

2.1. Governing equations and turbulence modeling

The two-dimensional unsteady incompressible RANS equations for a Newtonian fluid in dimensionless form, following the summation convention, are written as (in an inertial Cartesian reference frame):

$$\frac{\partial U_i}{\partial t} + U_j \frac{\partial U_i}{\partial x_j} = -\frac{\partial p}{\partial x_i} + \frac{1}{Re} \nabla^2 U_i - \frac{\partial \langle u_i u_j \rangle}{\partial x_j} \quad (1)$$

and the continuity equation:

$$\frac{\partial U_i}{\partial x_i} = 0, \quad (2)$$

where $i, j = 1, 2$.

Here the symbol $\langle \rangle$ denotes the temporal mean value of the contained variable, the lowercase letter u_i the velocity fluctuating components, the uppercase letter U_i the mean velocity components and p denotes the mean static pressure. All variables are expressed in a non-dimensional form with characteristic length scale the diameter of the cylinder D and velocity scale the free-stream velocity U_∞ . The main dimensionless parameters governing this problem are the Reynolds number $Re = \frac{U_\infty D}{\nu}$ and the rotational rate $\alpha = \frac{U_\theta}{U_\infty}$, where ν is the kinematic viscosity and U_θ the circumferential velocity of the cylinder.

The force and moment coefficients are non-dimensionalized by the cylinder's diameter and the dynamic pressure. Integration of the pressure and viscous forces along the surface gives the total magnitude of drag and lift. The torque coefficient is computed based on the lever-arm of every force acting onto the cylinder's surface. Respectively, they are defined as follows:

$$C_d = \frac{\int_{S_{cyl}} (\tilde{t}n) \cdot \underline{x} ds}{0.5 \rho U_\infty^2 D}, \quad (3)$$

$$C_l = \frac{\int_{S_{cyl}} (\tilde{t}n) \cdot \underline{y} ds}{0.5 \rho U_\infty^2 D}, \quad (4)$$

$$C_{m/(0,0)} = \frac{\int_{S_{cyl}} \|(\tilde{t}n) \times \underline{r}\| ds}{0.5 \rho U_\infty^2 D^2}, \quad (5)$$

where n is the vector normal to the surface, $\tilde{t}$ is the total stress tensor given by $\tilde{t} = -pI + \mu(\nabla \underline{u} + (\nabla \underline{u})^T)$, and $\underline{x}$, $\underline{y}$ and $\underline{r}$ are the x , y and radial position vectors, respectively. The center of rotation lies at $(0, 0)$ and it is the origin of all vectors.

After careful evaluation of selected RANS turbulence models and comparisons with available LES results, it was decided to adopt the standard $k-\varepsilon$ model with enhanced wall treatment. The latter model does not incorporate terms, which account for the rotational effects and there exists a known deficiency of the model to predict accurately the turbulence kinetic energy close to the stagnation area. However, the primary goal of this study is the computation of the mean loads and the $k-\varepsilon$ model with enhanced wall treatment, is superior over other models which account for rotational effects, for calculation of the mean pressure coefficient.

Special attention is given to the treatment of the near-wall region, which is important for the type of flow considered. The $k-\varepsilon$ model is by derivation a high- Re model to be combined with the wall-function approach [30]. In the present work, an enhanced wall treatment is utilized for making valid the $k-\varepsilon$ through the viscous-dominated region. This is accomplished by a two-layer model, where the turbulence viscosity and dissipation rate are computed in the near-wall cells. In the viscous-dominated near-wall region the one-equation model of Wolfshtein [31] is adopted. In this model the turbulence viscosity is computed based on the equation $\mu_{tv} = \rho \nu c (1 - e^{-Re_y/d})$, where c and d are constants and $Re_y = \frac{\rho y \sqrt{k}}{\mu}$. The smallest

length scales are computed based on the correlations of Chen and Patel [14] and their values are used to compute the dissipation rate based on the well-known relationship, $\varepsilon = \frac{k^{3/2}}{l_e}$, where l_e are the length scales in the viscous-wall region based on Patel; and k the turbulence kinetic energy, obtained from the mean flow kinetic energy equation (7). Finally the turbulent viscosity is blended with the high Re-number μ_{th} specified at the outer region (see Section 3.4.1). After careful examination of the resulting flow-field, it appears that this is an adequate treatment of the viscous sub-layer.

2.2. Computational grid

The grid used for these simulations is based on Padrino's grid [18], which is an O-type mesh. The latter type is proved to be advantageous for rotating surfaces. The C-type mesh [25] leads to a solution singular in the base point, when the cylinder has a non-zero circumferential velocity.

The grid is non-uniform. The hyperbolic tangent function is used to discretize the computational domain. Near the cylinder, the cell layers are very close to the surface, and the size of the cells gradually increases as the distance from the cylinder's boundary becomes longer. The grid is very fine and for the majority of the cases the y^+ (wall) value is on average close to 1.

The boundaries and the dimensions of the domain are illustrated in Fig. 1. Two symmetry boundary conditions are imposed far away from the cylinder (two straight lines near the poles of the circular grid). A velocity inlet condition is prescribed to the left region of the domain and a zero stream-wise gradient condition to the outflow region. Table 1 summarizes all the mathematical expressions of the boundary conditions. Along the cylinder's boundary, constant circumferential velocity $U_\theta = \alpha U_\infty$, consistent with counter-clockwise rotation, and zero radial velocity $U_r = 0$ are applied. The velocity inlet profile is preserved uniform and it is assumed that the cylinder's rotation does not have any impact on the inlet condition, based on the disturbance study of Kang et al. [25].

The position of the outer boundaries should not affect the pressure distribution across the cylinder's surface. Mittal and Kumar [33] and Padrino and Joseph [18] stated $75D$ as the distance between the cylinder's center and the outer boundary, after which, the force and moment coefficients do not change. A distance of $100D$ was selected in this study.

2.3. Numerical simulation

The numerical simulations presented in this work, are integrated in the framework of the commercial fluid dynamics package FLUENT 6.3. The elliptic partial differential equations are solved by the finite-volume method. The system of the algebraic equations is solved in a coupled iterative manner. Therefore, the velocity and the pressure fields are updated together. This is a coupled pressure–velocity procedure, which is very efficient and converges faster than the classical segregated approaches for well-behaved problems. The fully implicit coupling is achieved through an implicit discretization of the

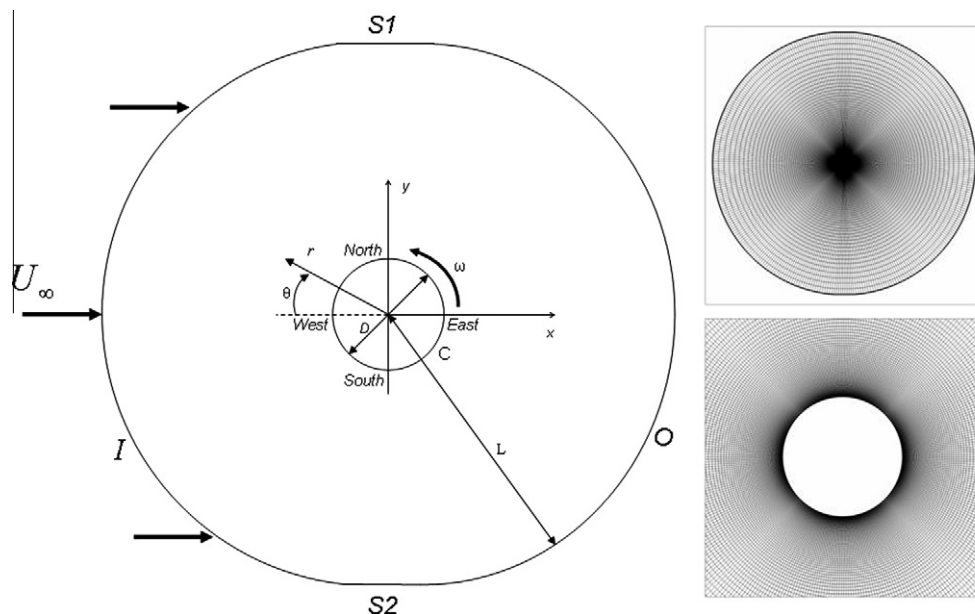


Fig. 1. Computational domain and system of reference for the numerical simulations. The boundary conditions are prescribed corresponding to the boundaries: I – inflow, $S1$, $S2$ – symmetry, O – outflow and C – constant circumferential velocity. The azimuth angle θ is referenced in a clockwise system, while the cylinder rotates in the counter-clockwise direction. A panoramic and a close-up view of the implemented structured grid are also presented in this plot.

Table 1

Mathematical expressions for every solved variable at the boundaries of the domain.

Boundary	Variable				
	U	V	k	ε	p
C	$-aU_\infty \frac{y}{\sqrt{x^2+y^2}}$	$aU_\infty \frac{y}{\sqrt{x^2+y^2}}$	0	$\nu \frac{\partial^2 k}{\partial n^2}$	$\frac{\partial p}{\partial n} = \mu \frac{\partial^2 U}{\partial n^2}$
O	$n \cdot \nabla U = 0$	$n \cdot \nabla V = 0$	$n \cdot \nabla k = 0$	$n \cdot \nabla \varepsilon = 0$	$n \cdot \nabla p = 0$
I	U_∞	0	$0.0096 \cdot U_\infty^2$	$1.102 \cdot 10^{-3} \cdot \frac{U_\infty^3}{D}$	$\frac{\partial p}{\partial n} = 0$
S1, S2	$\frac{\partial U}{\partial n} = 0$	0	$\frac{\partial k}{\partial n} = 0$	$\frac{\partial \varepsilon}{\partial n} = 0$	$\frac{\partial p}{\partial n} = 0$

pressure gradient terms in the momentum equations, and an implicit discretization of the mass flux, including the Rhie–Chow pressure dissipation terms. The above procedure is linked with an algebraic multigrid method and the set of equations is solved by a point or block Gauss-Seidel technique. The third-order (Van Leer) MUSCL scheme is selected to discretize the convective terms of the governing equations, while the second-order central scheme discretizes the diffusion terms. Rhie and Chow [29] interpolation for the computation of the mass fluxes in the continuity equation, is performed, to avoid checkerboarding for the implemented co-located grids. The process does not include pseudo-time terms and therefore the equations are solved implicitly (direct discretization and formulation of the algebraic system).

When solving the equations in their unsteady form, a second-order discretization scheme is utilized to integrate the equations in time. This scheme is implicit and the solving process possesses numerical stability. The time step is chosen based on the dimensionless step of $d\tau = 0.004$ (with time scale equal to $\frac{D}{U_\infty}$), according to the high Re studies of Catalano et al. [23] and Breuer [22].

3. Results and discussion

Three-dimensional RANS computations reported in the literature for cylinder flows, possess a two dimensional character (see [26]). Therefore, the present results might also be representative of 3-D RANS computations. Although, the RANS-URANS approach could not capture the 3-D features of the wake, the transition effects and the large-scale turbulent motions, it is known that in terms of the exerted loads acting on the cylinder's surface it is reliable and accurate. This is verified in the same study of Elmiligui et al. [26]. In the latter research article, it was also shown that for turbulent separated flows the k – ε is accurate enough for rotating cylinder flows with ratios a up to 1.

In this section, numerical results are presented at three Reynolds numbers and twelve rotational rates. Before discussing the results, a thorough validation of the numerical set-up is performed for selected published cases. First, the results from an unsteady run at $Re = 5 \times 10^5$ and $a = 2$, are presented. The reason for performing this unsteady computation is to verify that the vortex shedding is suppressed in our simulations, thus the steady-state runs produce valid and reliable results. Based on

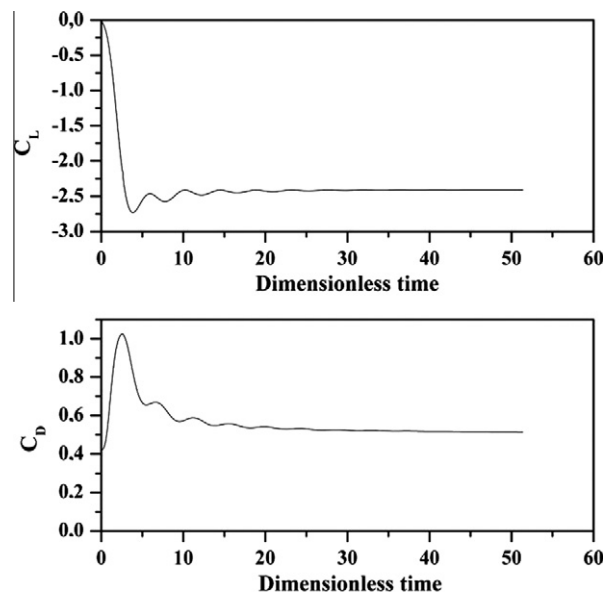


Fig. 2. Time variation of the lift and drag coefficients at $Re = 5 \times 10^5$ and $a = 2.0$. The flow becomes steady-state after a short dimensionless time. The flow-field is initialized with the free-stream inflow condition and no artificial perturbation is applied.

Table 2

Details for the meshes used in the grid-independency study. R_t^1 denotes the minimum radial distance between two successive grid points and N^C are the nodes, which are distributed onto the cylinder's surface.

Mesh	Nodes	Cells	L/D	R_t^1/D	N^C
M1	12,400	12,276	100	0.005	124
M2	86,840	86,506	100	0.0001	334
M3	160,640	160,138	100	0.00005	502

the study of Karabelas [24], the flow reaches steady state at $a = 2$ and at high sub-critical Re . Nevertheless, the present flow is supercritical and it has to be verified that the wake instability is no longer observed. Fig. 2 illustrates drag and lift coefficients plotted against dimensionless time. After a short dimensionless time, the mean flow ceases to be periodic and reaches steady state. Based on the literature [24] and [13], it is stated that at higher rotational rates and Re numbers, the turbulence characteristics of the flow damp the unsteadiness and any associated phenomena (e.g. vortex shedding). Therefore, it is considered that the flow becomes steady at rotational rates greater than $a = 2$ and Re numbers greater than 5×10^5 .

3.1. Validation of the computational code

Three different grids with structured topology following the O-mesh type of Padrino and Joseph [18] were used to ensure grid-independency of the results. Table 2 summarizes relevant details for every mesh. The pressure and skin-friction coef-

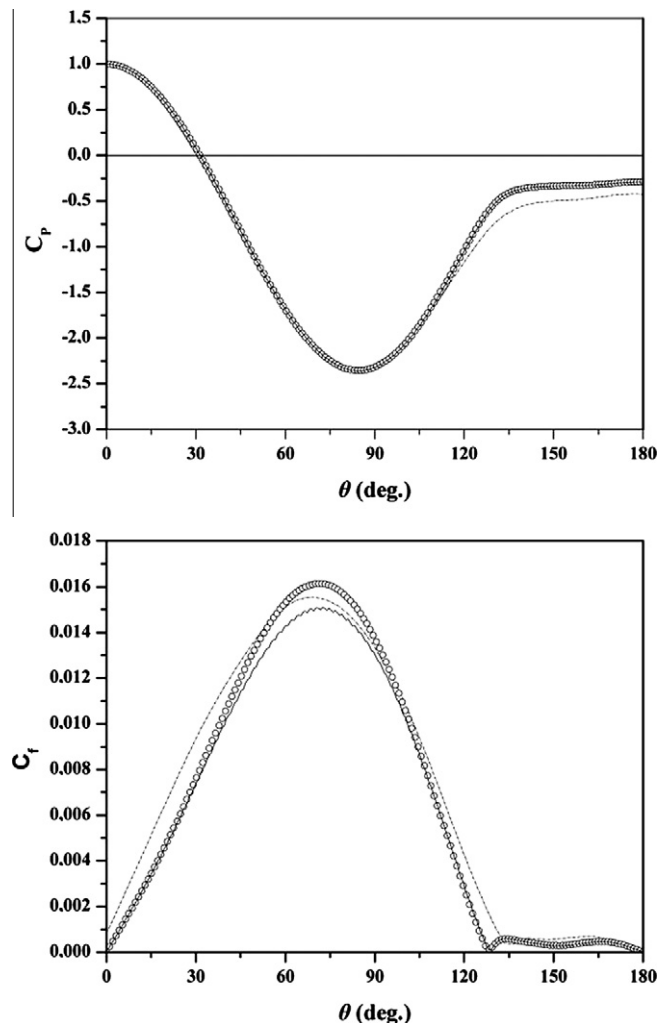


Fig. 3. Pressure and skin friction coefficients at the surface of the cylinder plotted for mesh I (dash-dotted line), mesh II (o) and mesh III (solid line). The rotational rate is zero and $Re = 10^6$.

ficients are compared at $Re = 10^6$. In this case, the cylinder translates (no rolling). Pressure coefficient in these simulations is defined as:

$$C_p = \frac{p - p_\infty}{0.5\rho U_\infty^2}, \quad (6)$$

where p_∞ represents the mean free-stream pressure at the inlet of the computational domain. Fig. 3 shows the pressure coefficient and skin friction curves for each of the three meshes at $Re = 10^6$. The results of meshes II and III are almost identical. In order to validate the present results, it was decided to compare them with available results in the literature. Comparisons with LES and experimental measurements for supercritical Re numbers ($Re > 3 \times 10^5$) from the open literature, demonstrate the validity of the numerical code.

Numerical results of Breuer [22] and Catalano et al. [23], who performed LES simulations for the non-rotating case, are chosen for comparison. Also relevant experimental results are available at Re near 10^6 . The Unsteady RANS equations are solved, since the flow is time dependent when $\alpha = 0$. Based on Kang [12], the initial conditions could be chosen arbitrarily. In the present runs at $Re = 10^6$ and $Re = 1.4 \times 10^5$, in order to trigger the vortex shedding process, it is decided to impose over half of the domain a velocity equal to U_∞ and to leave the other half stationary. Then, the computed profiles of the mean pressure coefficient at $Re = 10^6$ and of the turbulent shear stress $\langle uv \rangle$ at $Re = 1.4 \times 10^5$, are compared with available LES and URANS data. Fig. 4a plots the mean pressure coefficient along one semi-circle and Fig. 4b the turbulent shear stress $\langle uv \rangle$ at $x = D$ downstream. It is observed from this graph, that the $k-\varepsilon$ model suffices to predict the mean characteristics of the flow. A steady run has also been conducted and it is shown that when the enhanced wall treatment is adopted, the results are very close to the mean values of the LES simulations, apart from a small region at the suction side (Fig. 4a). The one-equation model of Spalart–Allmaras (RANS run) has also been tested for this occasion. It is observed that its inability to relate the local length scale to the shear layer thickness produces discrepancies. Other turbulence models were also tested,

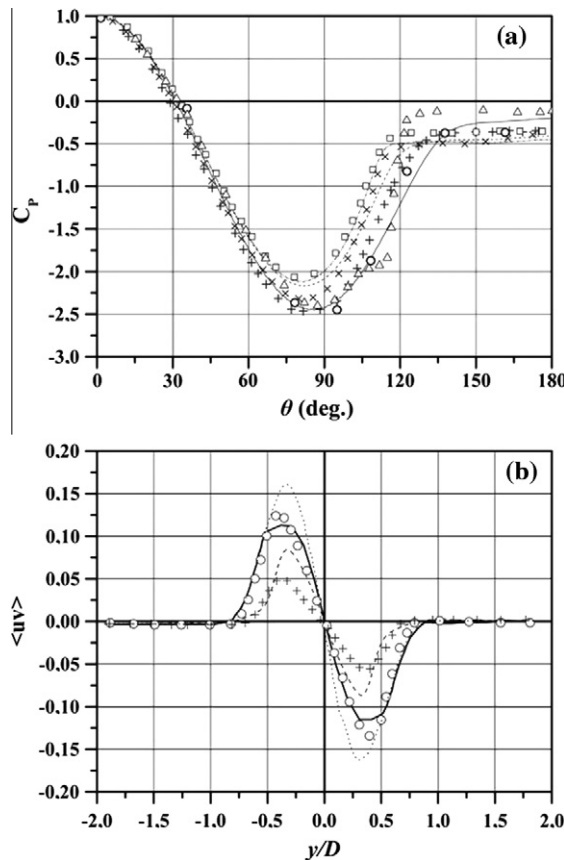


Fig. 4. (a) Pressure coefficient plotted as a function of the azimuth angle for one semi-circle of the cylinder's surface. Present computations: $k-\varepsilon$ (RANS) enhanced wall treatment model $Re = 10^6$ (solid line), $k-\varepsilon$ (URANS) standard wall function model $Re = 10^6$ (dash line), Spalart–Allmaras (RANS) $Re = 10^6$ (dash dot line), Catalano et al. LES results (+), Catalano et al. RANS results ($\square$), Catalano et al. URANS results ($\times$), experiment of Warschauer & Leene [34] at $Re = 1.26 \times 10^6$ (o) and experiment of Flachsbart in Zdravkovich [36] at $Re = 6.7 \times 10^5$ (Δ). (b) Reynolds shear stress plotted at $x/D = 1$ at $Re = 1.4 \times 10^5$. Experiment of Cantwell and Coles (solid curve), present $k-\varepsilon$ (URANS) (dash curve), Breuer [22] L.E.S.-fine grid with Smagorinsky model (dotted curve), Breuer [22] L.E.S.-coarse grid with dynamic model (o) and Breuer [22] L.E.S.-coarse grid with Smagorinsky model (+).

Table 3

Drag coefficient, Strouhal number and pressure base coefficient computed at $Re = 10^6$ by Catalano et al. [23] and measured by Shih et al. [35] and Zdravkovich [36]. The experimental and numerical data are compared to the present results.

	C_d	Strouhal number (St)	$C_{v,base}$
Present results	0.34	0.20	−0.30
LES, Catalano et al. [23]	0.31	0.28	−0.32
Experimental results, Shih et al. [35]	0.24	0.22	−0.33
Experimental results, Zdravkovich et al. [36]	0.17–0.40	0.18–0.50	–

such as the *RNG* $k-\varepsilon$ and the *SST* $k-\omega$ but they appeared to produce significant differences from the cited results. Therefore, for the present simulations the $k-\varepsilon$ model is adopted, as modified for resolving the flow-field through the viscous and buffer layers. Table 3 also summarizes the basic integral parameters (drag coefficient, Strouhal number and pressure base coefficient), computed in the present study and in studies of other researchers.

To further validate the code for the rotating case ($a > 0$), it was decided to conduct a separate study at $Re = 60,000$, where numerical and experimental data are available. Fig. 5a plots the lift coefficient as a function of the rotation rate a . The computed lift coefficient agrees quite well with the experimental measurements. A small discrepancy appears at medium rotation rates close to the value of 0.5. On the contrary, Aoki's computations greatly diverge from the experimental measurements at rates greater than 0.5 (maximum lift coefficient is 1.8 at $a = 0.78$). Drag coefficient is also plotted in Fig. 5b. It is observed, that the computed drag coefficient is over-predicted, when compared to the experimental and the numerical 3-d results. However, the present computations are two dimensional and higher values of the drag coefficient are normally expected. Travin et al. [28] in their detached eddy simulations, found even higher value ($C_d = 1.77$) from the present one for the non-rotating case, as it is clearly observed in Fig. 5b. Aoki's results agree better with the experimental measurements, but there is still an over-prediction in the computation of the drag force.

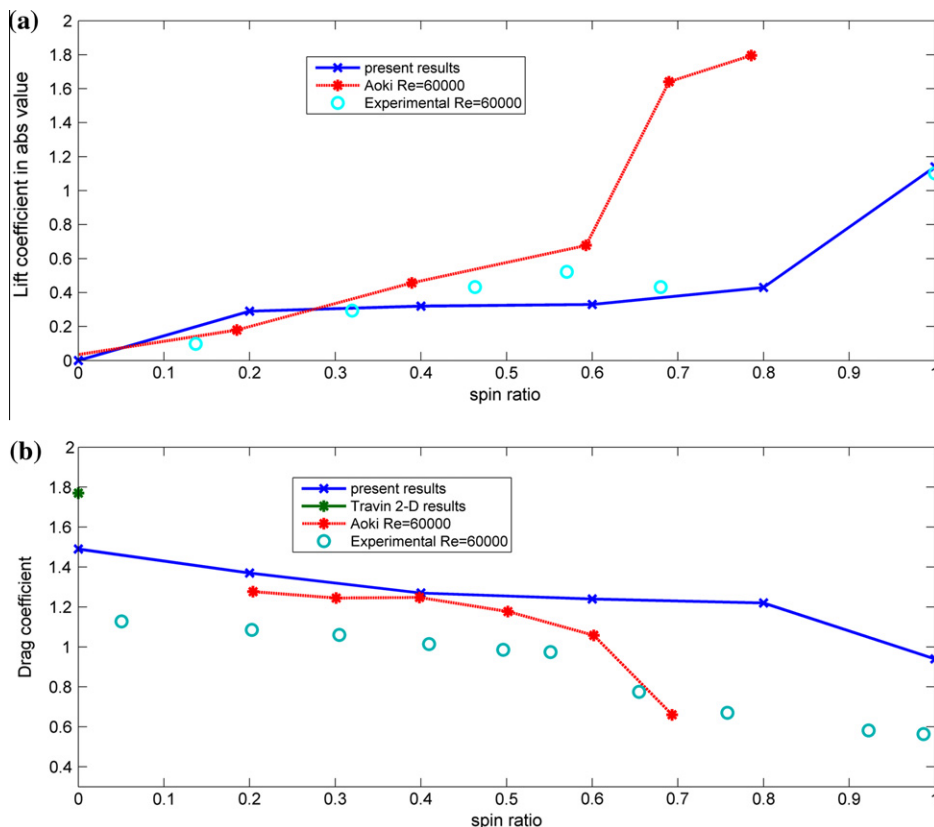


Fig. 5. (a) Lift coefficient plotted versus rotation rate. Present computations at $Re = 60,000$ (solid line with x), Aoki's computations cited in Elmiligui [26] at $Re = 60,000$ (solid line with +) and experimental measurements cited in Elmiligui [26] at $Re = 60,000$ (o). (b) Drag coefficient plotted versus rotation rate. Present computations at $Re = 60,000$ (solid line with x), Travin 2-D results (green circle) at $Re = 50,000$, Aoki computations cited in Elmiligui [26] at $Re = 60,000$ (solid line with +) and experimental measurements cited in Elmiligui [26] at $Re = 60,000$ (cyan circle). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

3.2. Two dimensional flow patterns at various Re and a

For the examined dimensionless rotational rates and Reynolds numbers the flow reaches steady state. In the laminar regime this is true under specific conditions. The α , Re should lie in a specific range of values. The supercritical flow patterns are quite different from the laminar ones. Fig. 6 presents supercritical streamline patterns compared with laminar ones at $Re = 200$ for some common examined cases (same rotational rate). It can be observed that for the laminar case when $a = 2$, there are two apparent stagnation points. The first (L1) is attributed to the collision of the rotating boundary layer with the free-stream flow, while the second (L2) to the creation of a strong vortex downstream of the cylinder, which is formed due to the strong vorticity gradient between the free-stream layer and the rotating fluid, moving upstream and towards the top of the cylinder. As the dimensionless rotational rate increases, the upstream stagnation point moves downstream until

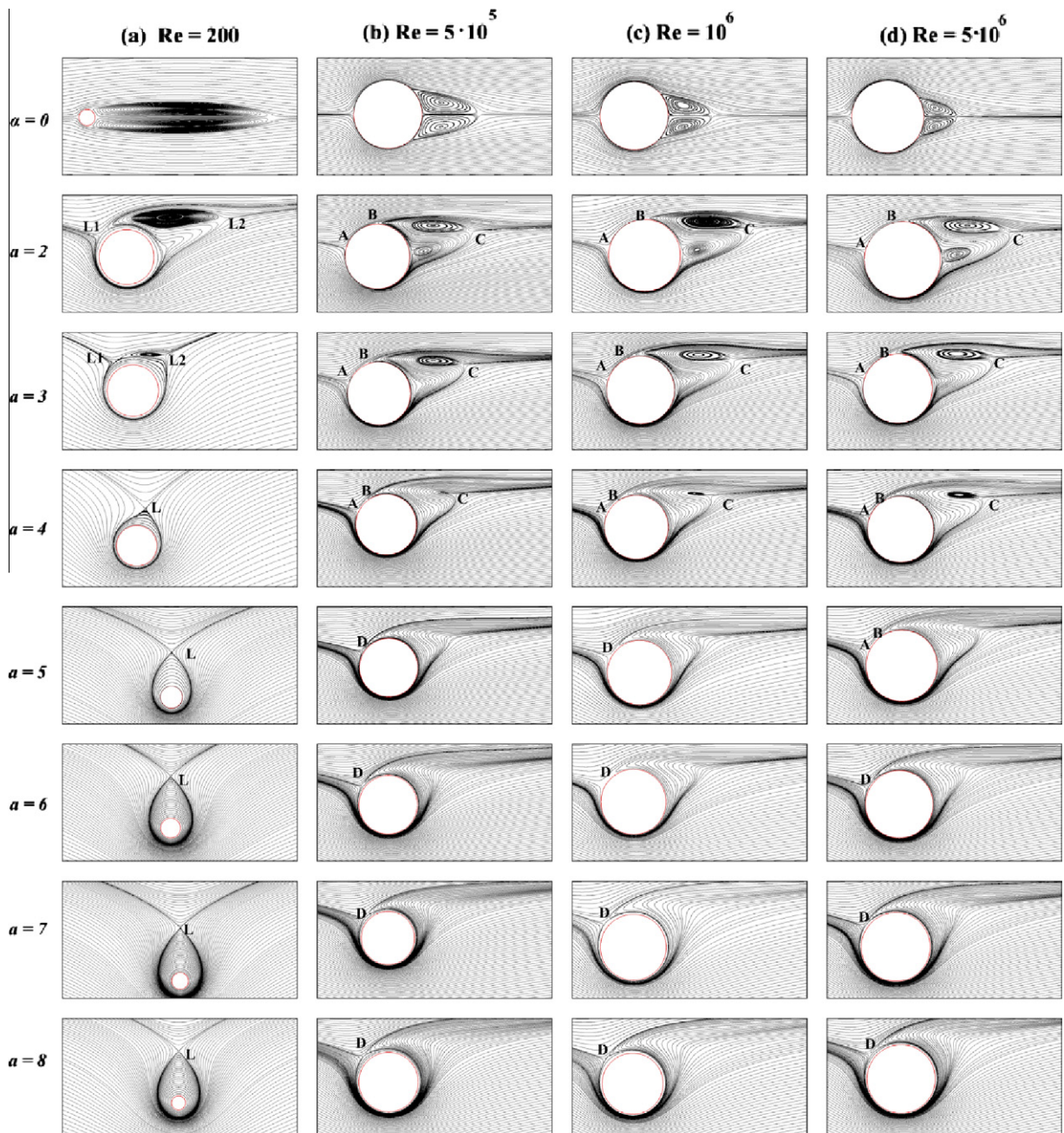


Fig. 6. Streamline patterns at $Re = 200, 5 \times 10^5, 10^6, 5 \times 10^6$ and rotational rates $a = 0, 2, 3, 4, 5, 6, 7$ and 8 . L1 and L2 stagnation points are apparent for low rotational rates (laminar flow), while A, B, C and Z are addressed for the super-critical Reynolds numbers.

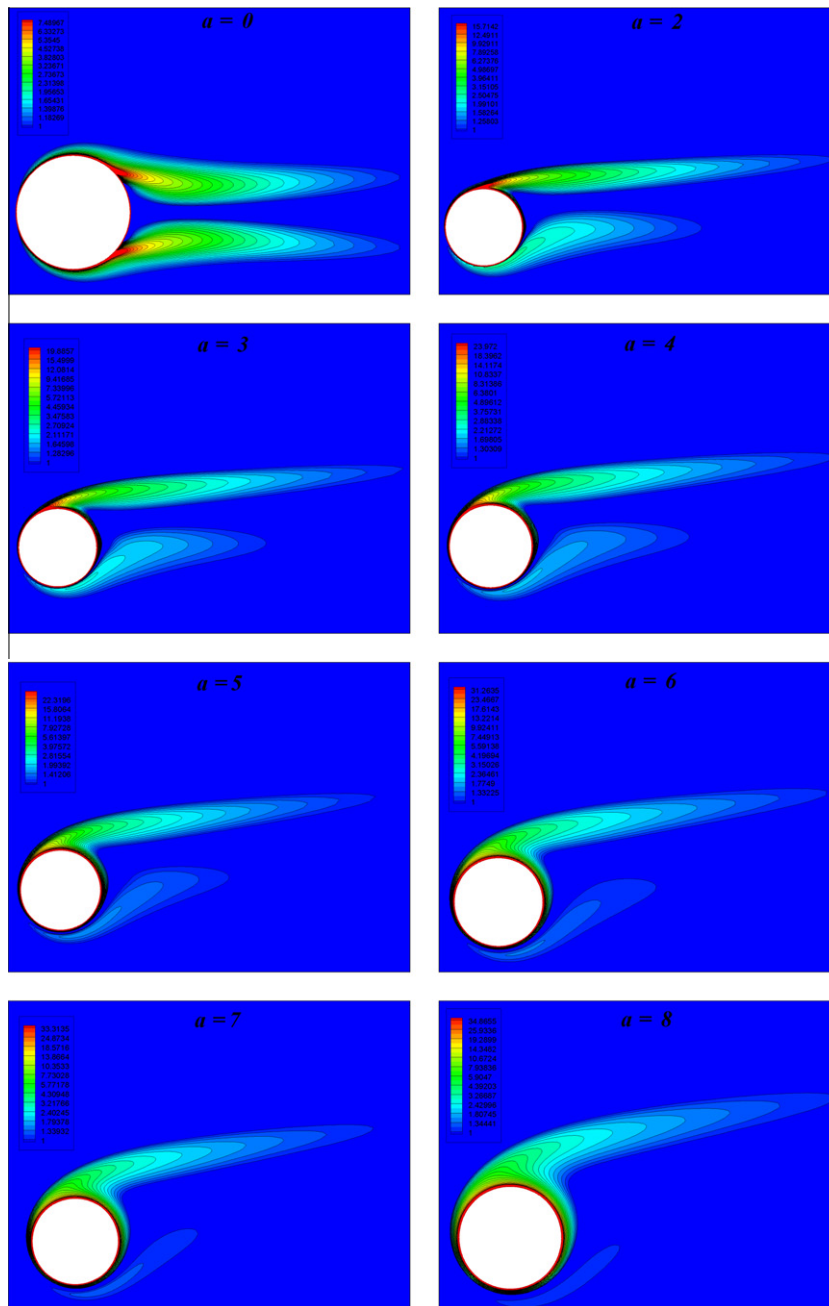


Fig. 7. Contours of dimensionless vorticity modulus, defined as $\bar{w} = \|\bar{\omega}\|D/U_\infty$ at $Re = 10^6$. The vorticity field is colored for values always less than 0.1% of the maximum vorticity.

$a = 4$. The downstream vortex begins to contract and fully collapses at $a = 4$. For greater a the flow becomes swirling with a single stagnation point L, which moves towards the outer region of the free-stream flow.

In supercritical regimes, different patterns are revealed. It can clearly be observed that the high Re patterns present another stagnation point (B) near the cylinder's surface, apart from that one (A) at the upstream area and another two (C), (Z) close to the downstream vortices. For $a = 2$ two vortex structures appear at the suction side at $\theta = 180^\circ$ and $\theta = 150^\circ$, approximately. For $a = 3$ the first vortex collapses. For greater rotational rates the flow deviates upstream and the second vortex is also suppressed (stagnation point (C) collapses). However, as far as it is strong, it lies in the same area. It can be observed that (A) and (B) come closer as the rotational rate increases. At $a = 5$ they merge and form a new single stagnation point (D). After $a = 5$ (D) moves outwards at higher radial positions, while its' azimuth position changes slightly.

The ability of the rotating fluid to remain attached to the cylinder is also related to the general characteristics of the supercritical boundary layers. However, at low a there is a separation area (in the sense of a downstream stable vortex), different from that one in the non-rotating case, but as the rotational rate increases the boundary layer separation vanishes and the flow is influenced by an intensely swirling velocity flow-field.

Generally it appears that the flow results are independent of Re . However, some distinctions are apparent for medium rotational rates. For low rotational rate $a = 2$ the flow patterns are almost similar. The vortex structure near the suction point is more slender for $Re = 5 \times 10^5$, while the main vortex is slightly wider for the latter Re compared to $Re = 5 \times 10^6$. There are also some differences in terms of vortex dimension for $a = 3$ and $a = 4$. At $Re = 5 \times 10^6$ the vortex downstream of the cylinder is thicker. For greater ratios, the stagnation points are sited at similar positions and the development of the swirling fluid layer is almost the same.

3.3. Vortical activity and force coefficients

The flow over a rotating cylinder exerts on it significant forces, with different magnitude and direction compared to the loads generated in the non-rotating case. One of the major topic analyzed in aerodynamics, is the generation of lift due to the Magnus effect [12]. The rotation also produces torque, which acts against the motion of the fluid. This torque is mainly the outcome of the viscous forces acting onto the cylinder's surface, as it will be shown in the present section.

When turbulence cannot longer be ignored, more intense concentrated vortical regions are observed, because of the appearance of turbulence shear stresses inside the boundary layer. The intensity of this vortical activity is strongly dependent on Re , since the boundary layer thickness is always subject to Re . In supercritical flows the main vortical activity is confined near to the surface of the cylinder and the rest of the domain possesses much lower vorticity. Fig. 7 plots contours of dimensionless vorticity's modulus, indicating fluid of low vorticity (values less than 0.1% of the maximum value are colored). This pattern is also representative of laminar flows, where the cylinder is exposed to high rotational rates without any wake instabilities.

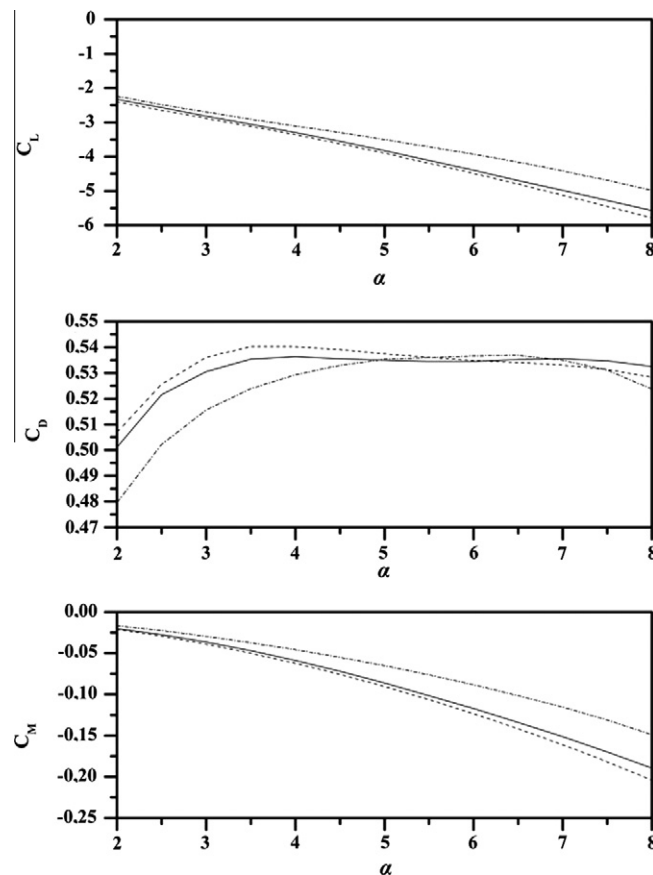


Fig. 8. Lift, drag and moment coefficients plotted versus dimensionless rotational rate at $Re = 5 \times 10^5$ (dash line), $Re = 10^6$ (solid line) and $Re = 5 \times 10^6$ (dash-dotted line).

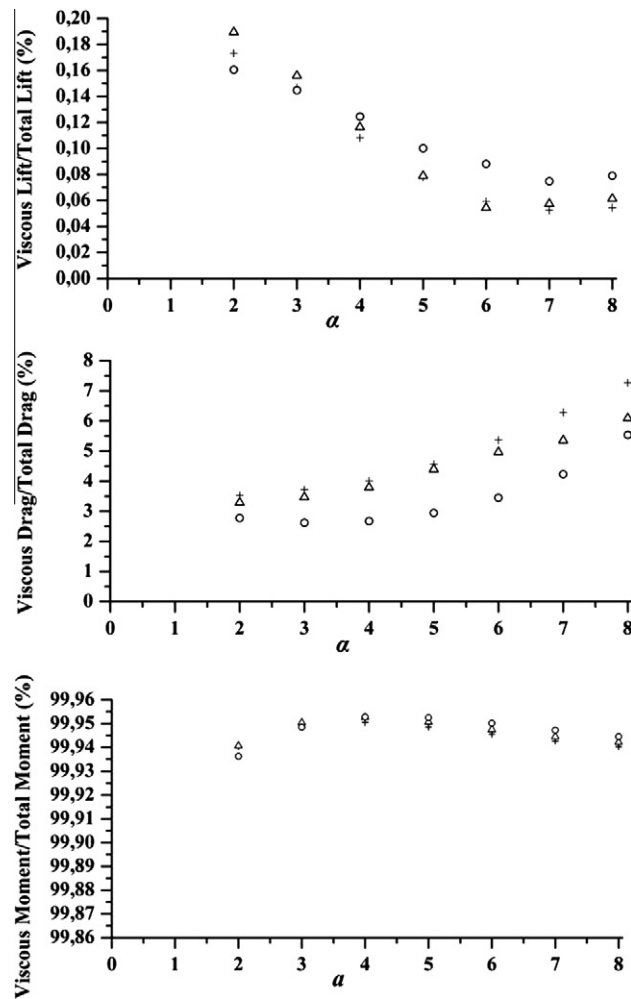


Fig. 9. Viscous drag, viscous lift and viscous moment divided with the total drag, lift and moment, respectively. The viscous contributions are plotted against the rotational rate at $Re = 5 \times 10^5$ (+), $Re = 10^6$ (Δ) and $Re = 5 \times 10^6$ (o).

The numerical results for the total lift, drag and torque coefficients are plotted in Fig. 8. The plots include also the trends for the ensemble of the examined Re . At $Re = 5 \times 10^5$, a slight increase in cross-stream force is observed compared to $Re = 10^6$, while the lift becomes smaller at $Re = 5 \times 10^6$. The cross-stream force is directed towards the negative direction of y -axis, thus a downward force is always produced. For constant Re the lift coefficient increases almost linearly. At $Re = 5 \times 10^6$ a mildly different behavior is exhibited, which might be due to the inability of the grid to resolve adequately the viscous wall region for rotational rates greater than 3. The contribution of the viscous component to the total lift is also included and is plotted in Fig. 9. Values of the viscous lift do not exceed 0.2%, therefore it can be considered negligible. In high rotational rates the viscous effects in lift terms are almost the same at $Re = 5 \times 10^5$, 10^6 and more intense at $Re = 5 \times 10^6$. However, at $a = 0$ (RANS run) viscous effects are more pronounced at $Re = 5 \times 10^5$, followed by $Re = 10^6$ and finally $Re = 5 \times 10^6$. At $a = 2$, the viscous lift force is maximum at $Re = 10^6$. In addition, the pressure and viscous lift forces counterbalance each other, since they are applied in opposite directions (viscous lift points towards positive y). Drag coefficient is also plotted in Fig. 8 and it can be observed from this plot, that drag varies similarly for all the examined cases, thus it increases with a and decreases with Re . The viscous drag force contributes differently than the viscous lift. Fig. 9 shows that the viscous drag varies in the range of 2.5–7% of the total drag and it increases with rotational rate. The pressure and viscous drag are both directed towards the x positive or free-stream direction. It is also clear from the same plot that there is an increase of the viscous drag contribution to the total drag as Re decreases. The drag and lift viscous forces produce torque. The pressure forces do not produce considerable torque, as it is shown in Fig. 9. The curves plotted in Fig. 8 reveal significant increase of the torque as a function of the rotational speed. This trend is clearly similar to the lift's one. The direction of the torque's vector points inwards and it resists the cylinder's rotational motion.

3.4. Turbulence analysis

For a comprehensive study of turbulence, the understanding of the contribution of every Reynolds stress-tensor component to the momentum and energy balances is essential. There exist multiple studies in the literature, analyzing the spatial spectrum for the turbulence kinetic energy (half of the Reynolds tensor's trace), thus the energy distribution across the different length scales. Two relevant studies are those of Singh's and Mittal's [20] and Breuer's [22] for the non-rotating case, but it appears from the literature that there is no relevant work on high-Re turbulent flow past rotating cylinders. In the present paper, we analyze the distribution of the kinetic energy and the shear-stress component starting from the viscous sub-layer and extending to the inertial sub-layer for multiple azimuth positions along the cylinder's boundary. Next, the contribution of the energy transfer terms to the total energy balance is evaluated. The budget of the turbulence kinetic energy is evaluated at multiple azimuth positions and dimensionless rotational rates a .

3.4.1. Reynolds stress tensor field

In statistically two dimensional flows $\langle uw \rangle = 0$, the Reynolds stress tensor takes the following form [27]:

$$R = \begin{pmatrix} \langle uu \rangle & \langle uv \rangle & 0 \\ \langle uv \rangle & \langle vv \rangle & 0 \\ 0 & 0 & \langle ww \rangle \end{pmatrix}. \quad (7)$$

The Reynolds normal stress components constitute the turbulence kinetic energy, which is equal to half of the tensor's trace, hence $k = 0.5 \cdot (\langle u^2 \rangle + \langle v^2 \rangle + \langle w^2 \rangle)$. The normal components in simple shear flows decay as the distance from the wall increases. However, this is not the case for a rotating cylinder, since very intense turbulent regions grow far away from the turbulent boundary layer's core. The shear component $\langle uv \rangle$, becomes high inside the turbulent boundary layer, where the production term is non negligible, as it will be shown in this section. However, far away from the wall it could be also significant.

Special attention is needed for the computation of ν_t through the viscous sublayer and the buffer region. The turbulence kinematic viscosity is given by: $\nu_t = 0.09 \frac{k^2}{\epsilon}$ [29,30]. However, this expression ceases to be valid through the viscous wall region. To extend through the inner layer, the equations of Wolfshtein [31] and Chen and Patel [14] are used for modifying the turbulence viscosity. For the smooth blending of the dissipation and turbulence viscosity profiles with the outer region, the following expression proposed by Jørgen [32] is applied:

$$\mu_t = \lambda \mu_{th} + (1 - \lambda) \mu_{tv} \quad (8)$$

with λ being a function of Re_y and it is defined in such a way that it is equal to 1.0 far from the wall and 0.0 very close to it.

The Reynolds shear stress could be computed from the production term, by the following expression, which is valid only through the turbulent boundary layer:

$$\langle uv \rangle = P \cdot \left(\frac{\partial V'}{\partial n} \right)^{-1},$$

where V' is the mean velocity component parallel to the cylinder's surface.

Fig. 10 plots the contours of the turbulence kinetic energy k for rotational rates $a = 0$ (RANS run) to $a = 8$ in the neighborhood of the cylinder's surface. It can be observed that when the cylinder rotates, the region of maximum k is shifted downstream and towards the south area (acceleration region). When the dimensionless rotational rate reaches the value of 6.0 then this region is indicated approximately at azimuth angle $\theta = 60^\circ$. Based on the streamline patterns plotted in Fig. 6, the stagnation point is located in the latter area, where the vorticity gets maximum. However, this trend is not observed for the lower rotational rates $a = 2$ and $a = 4$, thus the turbulence kinetic energy and the vorticity do not present their maxima in the same region. This may be attributed to the distribution of the $\frac{\partial V'}{\partial n}$ term. For medium dimensionless rotational rates, the term $\frac{\partial V'}{\partial n}$ is greater at south, although vorticity is less there than at the north area. Nevertheless, the vorticity distribution is not always analogous to that one of $\frac{\partial V'}{\partial n}$, because vorticity, expressed in a shifted reference frame (t, n) , where t denotes the tangential and n the radial direction, is equal to:

$$\vec{\omega} = \left(\frac{\partial V'}{\partial n} - \frac{\partial N'}{\partial t} \right) \vec{z},$$

where N' denotes the mean radial velocity component. For very high a the term $\frac{\partial V'}{\partial n}$ is always higher in the north region.

The profiles of k for various rotational rates are plotted in Fig. 11. A steep increase of energy is observed through the viscous sub-layer, followed by a gradual decay. Careful inspection of these diagrams, show that the decay is slower as the rotational rate increases. Other results derived from this plot could be summarized as followed: (1) The least decay rate is observed in the suction and the southeast regions ($\theta = 180^\circ$ and $\theta = 225^\circ$, respectively). (2) The kinetic energy diminishes at $\frac{y}{D} \geq \{0.4-0.7\}$ at azimuth angles $\theta = 0^\circ, 45^\circ, 90^\circ, 135^\circ, 270^\circ$ and 315° and $\frac{y}{D} > 1.0$ at $\theta = 180^\circ$ and 225° . (3) The profiles of k vary in a different way, always dependent on the azimuth angle, thus the flow does not become self-similar in the circumferential direction and 4. The maximum kinetic energy is observed for $a = 8$ and $\theta = 45^\circ$.

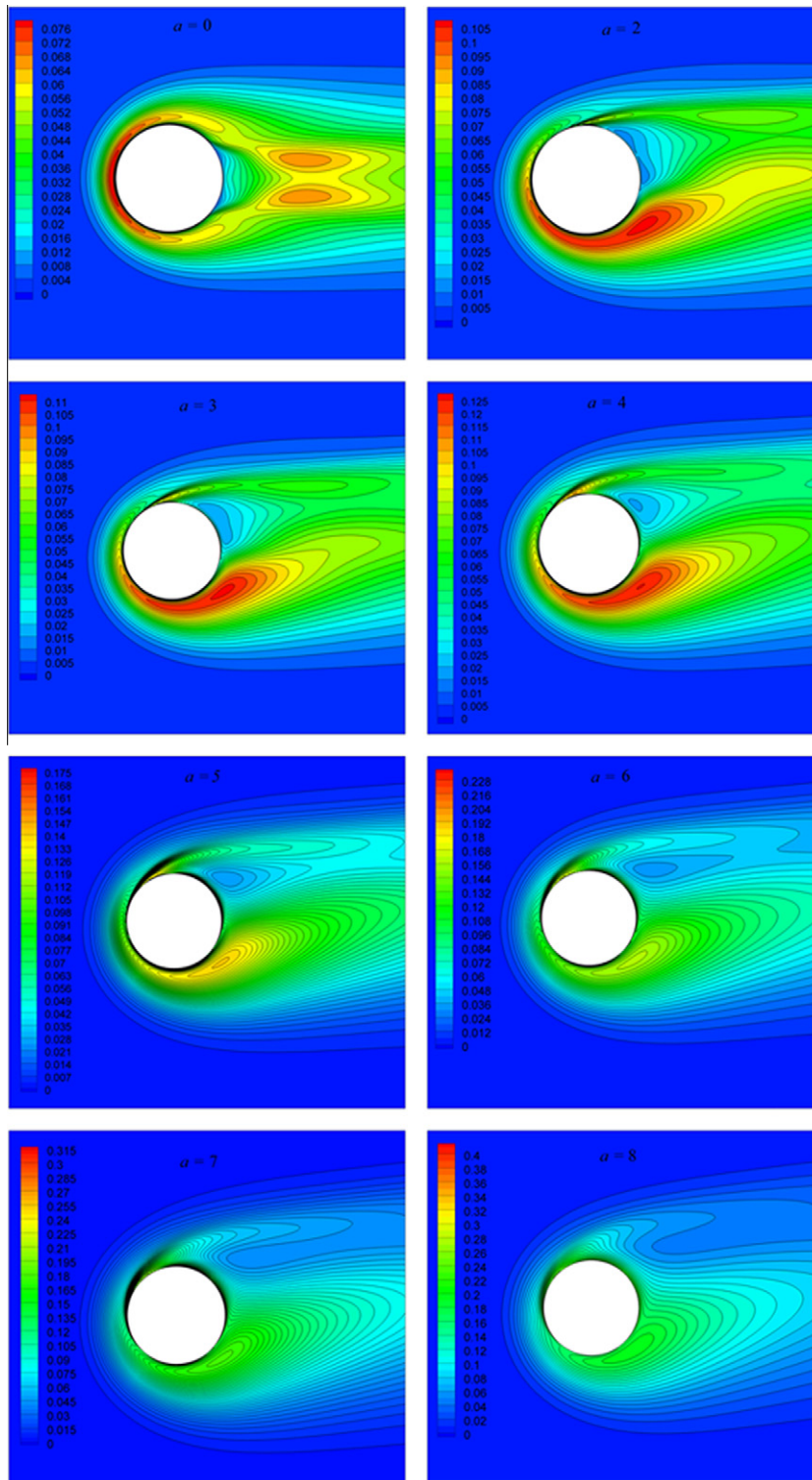


Fig. 10. Turbulence kinetic energy contours normalized by U_∞^2 , at $Re = 10^6$ and rotational rates $a = 0, 2, 3, 4, 5, 6, 7$ and 8 . The area of maximum energy is displaced azimuthally, as the rotational rate increases.

The Reynolds shear stress is also plotted in Fig. 12 for four rotational rates and two azimuth positions, which are the north and the south pole, respectively. It can be observed from these plots that the turbulence shear stress is more intense in the south pole. Also it becomes negative for $a = 0$ and $a = 2$ at specific portions of the boundary layer. In more details, for $a = 0$ it

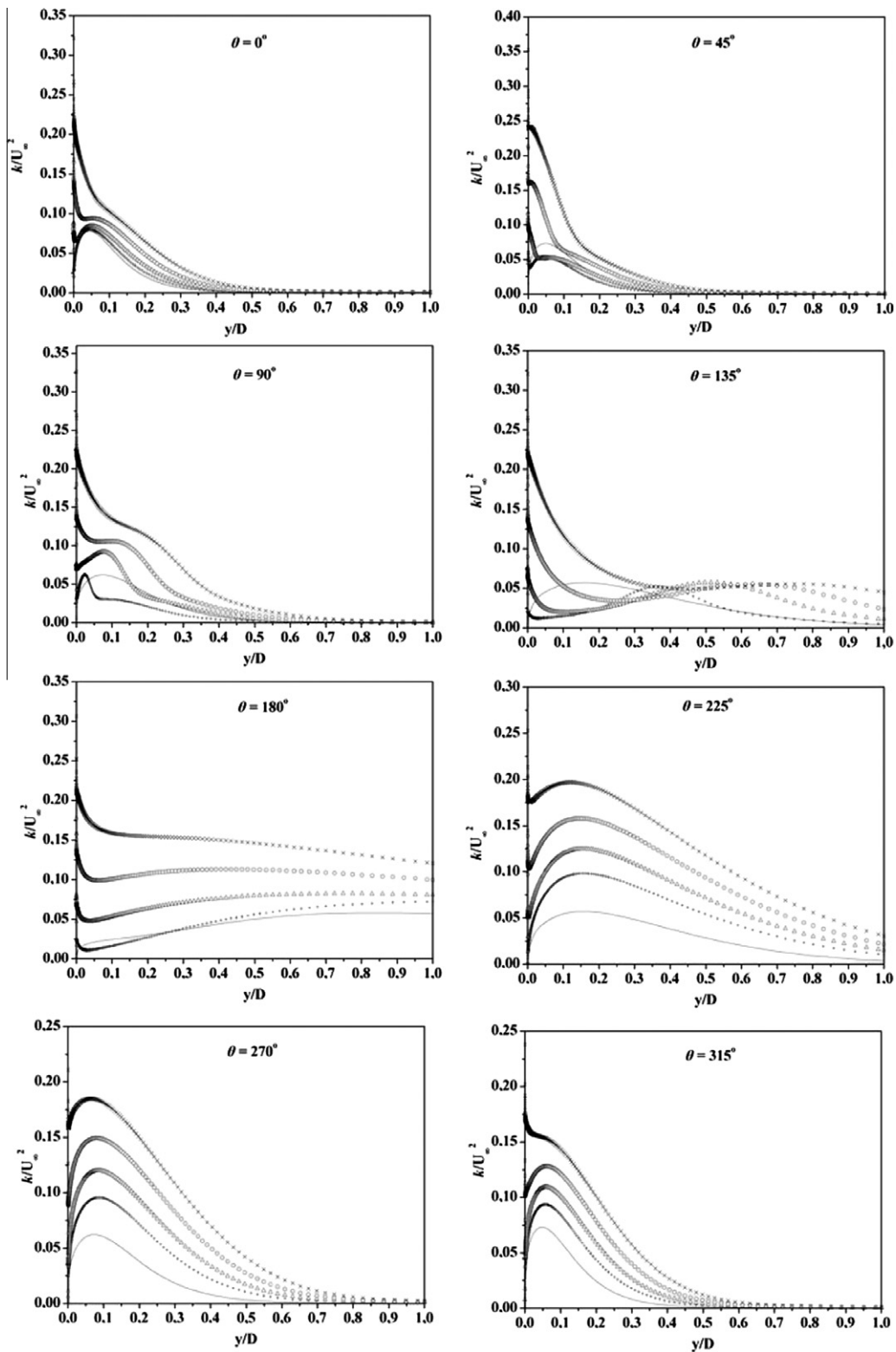


Fig. 11. Turbulence kinetic energy profiles normalized by U_∞^2 at $Re = 10^6$ at various azimuth angles, plotted against the dimensionless position y/D . The rotational rates are denoted as: $a = 0$ (solid line), $a = 2$ (+), $a = 4$ (Δ), $a = 6$ (o) and $a = 8$ (x).

changes sign (the term $\frac{\partial v}{\partial n}$ changes sign) and after $\frac{y}{D} > 0.025$ it becomes positive. For $a = 2$ it remains negative when $\frac{y}{D} > 0.01$ for the north pole. In the south pole the stress is always positive at non-zero rotational rates.

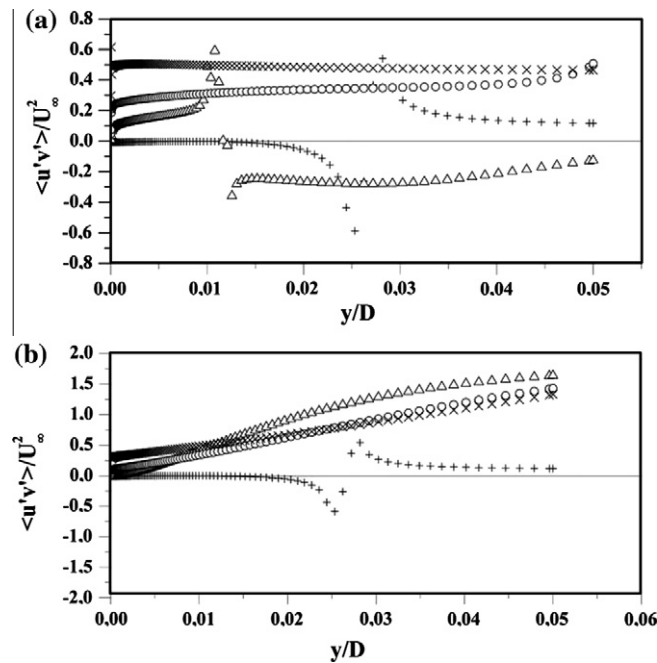


Fig. 12. Turbulent shear stress normalized by U_∞^2 , at $Re = 10^6$ and $a = 0$ (+), $a = 2$ (Δ), $a = 4$ (o) and $a = 6$ (x), plotted against y/D at the south (bottom) and north (top) poles.

3.4.2. Budget of the turbulence kinetic energy

In the majority of turbulent flows, the most competitive terms in the energy balance, are the production P and the dissipation ε , as defined previously. However, the turbulence transport term G , could play a significant role in some portions of the boundary layer. On the contrary, the turbulence convection term is almost negligible, because it is dependent on the mean velocity components and the k gradients, where the former are low through the boundary layer. It can be observed from Fig. 13, that the budget of the turbulence kinetic energy does not differ significantly from that one of channel flow. Four azimuth positions have been selected at $\theta = 45^\circ, 90^\circ, 225^\circ$ and 270° to plot the profiles of each energy term for dimensionless rotational rates from $a = 0$ up to $a = 6$. It is evident from the ensemble of the diagrams that the dissipation and production terms are the most important contributors to the energy transfer. In most cases, production is less than dissipation through the viscous sub-layer, since the dissipation term is usually high enough close to the wall, thus equal to the pseudo-dissipation term $\bar{\varepsilon} = \nu \frac{d^2 k}{dn^2}$. In the aforementioned region, the turbulence transport term constitutes a source of energy, therefore balancing the energy transfer rate. In the beginning of the buffer layer, the latter term becomes a sink of energy and eventually decays after a short distance or alternates sign for one more time. Generally, in the log-law region the foremost terms are well compensated, giving negligible energy transport and convection terms. However, in some cases the dissipation decays slower than production and the transport contributes as a source of energy. There exists also a case at $\theta = 225^\circ$ and $a = 0$, where the convection term contributes as a main sink of energy comparable to the dissipation term. At this θ , the energy terms and their decay rate are minimal.

Production and dissipation terms grow steeply as a function of the rotational rate for all azimuth angles, justifying the corresponding increase of the turbulence kinetic energy, illustrated in Fig. 13. Also the convection and transport terms follow the same trend. It seems that the energy terms possess an almost similar behavior in terms of the azimuth position. That is, even when their values are not the same azimuthally, the trend is analogous.

4. Conclusions

Uniform flow past a rotating cylinder is investigated via RANS and URANS simulations, for several rotational rates ($a = 0$ to $a = 8$) and supercritical $Re = 5 \times 10^5, 10^6$ and 5×10^6 . The finite volume method is implemented to solve the incompressible RANS equations. An implicit coupled solving technique is applied for the present simulations, which preserves the primitive variable formulation of the equations and solves the Poisson equation for the pressure correction and the momentum equations for the velocities. The mean flow is found to reach steady state for the ensemble of the examined cases.

Flow patterns for turbulent high Re flows are affected by a in a different manner than for laminar flows. In supercritical flows, the patterns are almost the same for $a > 4$, whereas in laminar flows for $a > 4$ the stagnation point is shifted continuously towards the outer region (Fig. 6). However, in both regimes for high spin rates, there exist only one stagnation point and all the vortices, formed at lower a , collapse. Magnus effects are not as much pronounced as in the laminar state. Also, in

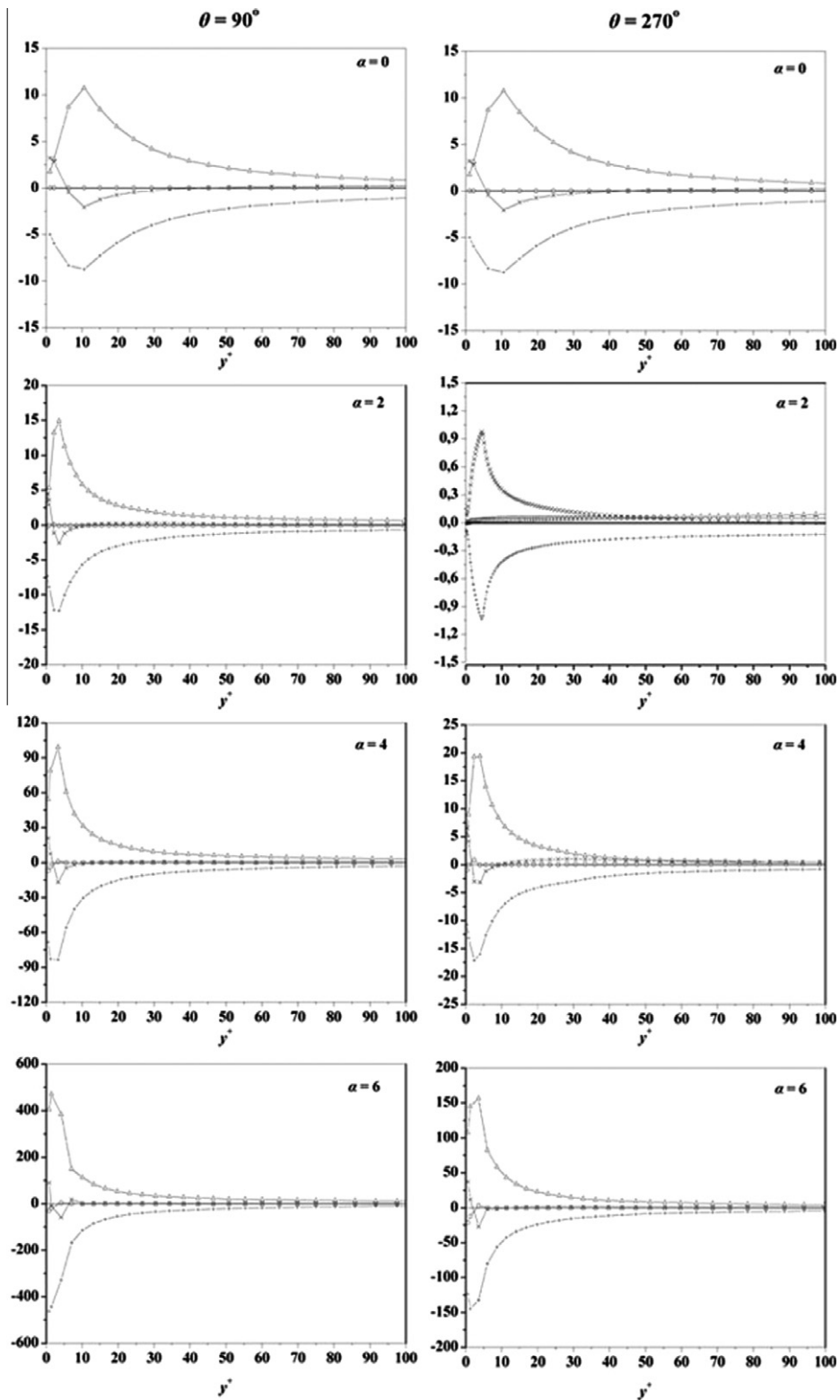


Fig. 13. Budget of the turbulence kinetic energy at $Re = 10^6$ and $\alpha = 0, 2, 4, 6$ at various azimuth angles plotted against the local dimensionless y^+ value. The energy terms are normalized by $\frac{U_b^3}{D}$ and they are denoted as: dissipation (solid line with +), production (solid line with Δ), convection (solid line with o) and transport (solid line with x).

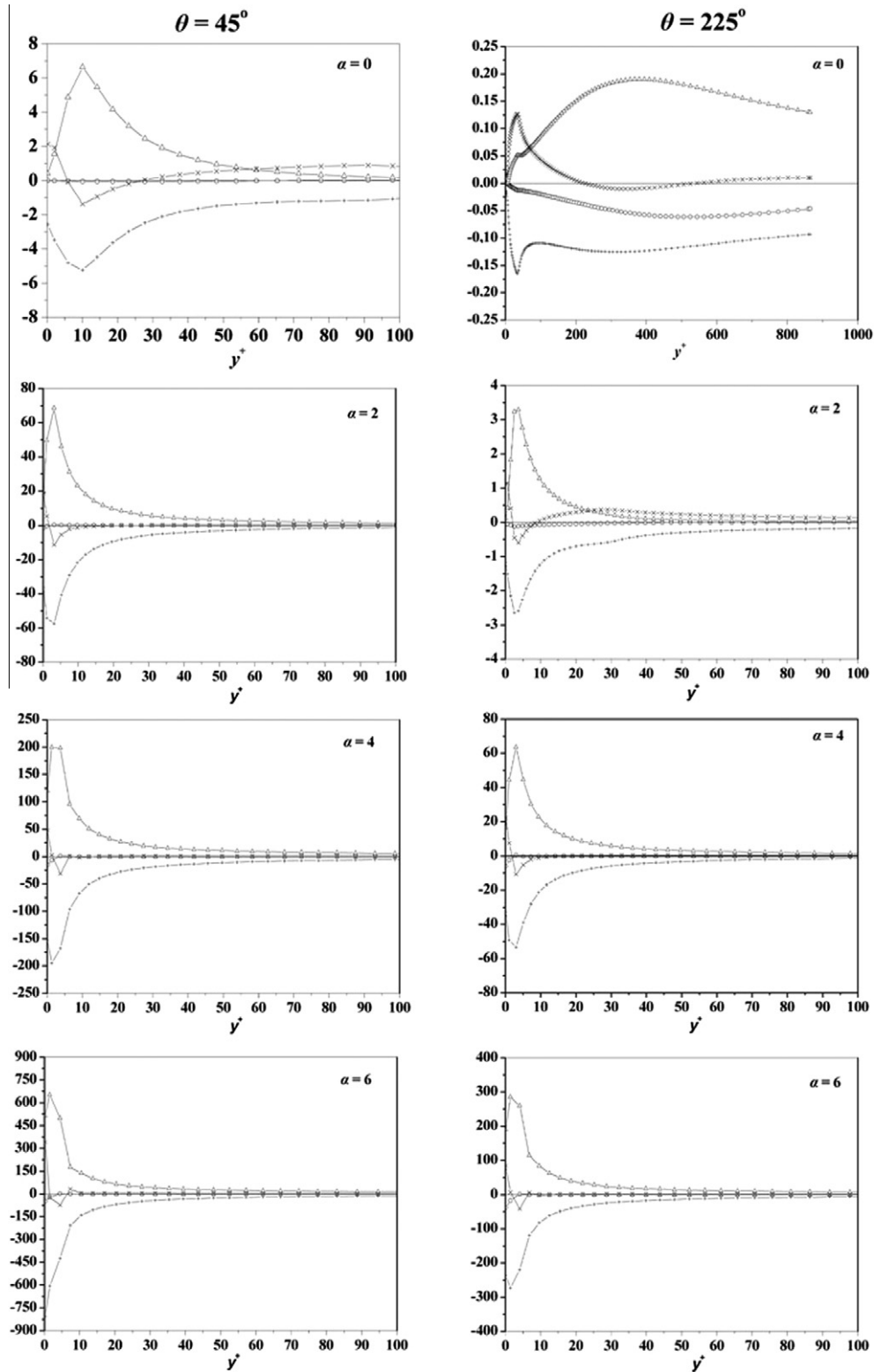


Fig. 13 (continued)

the examined Re range, it is found that the loads are slightly affected by the Re number (no more than 10% for lift and drag). Lift increases linearly with a and the drag force increases up to $a = 4$, where it reaches a plateau and eventually decreases.

Viscous force is important only in drag terms, where it could reach 7% of the total drag. Nevertheless, viscous lift and viscous drag are the foremost contributors to the torque production.

Turbulence kinetic energy gets maximum near the south pole at $\theta = 270^\circ$ for a up to 5. At higher spin rates k peaks near the stagnation point, where the vorticity is also maximum. Generally k decays at least two to three times slower in the suction side than in the pressure side. Turbulence dissipation rate, kinetic energy and shear stress are found to increase non-linearly with dimensionless rotational rate. The latter one is more intense in the south pole and could become negative at $a = 0$ at both poles and for $a = 2$ at the north pole. Production and dissipation of turbulence kinetic energy are well compensated in the inner layer of the log-law region. Through the viscous sub-layer, dissipation prevails, giving rise to the transport term, which acts as a source. In most of the examined cases, the decay rate of production was slightly higher than dissipation rate in the outer layer. All the energy rate terms increase steeply with dimensionless rotational rate a and decay slower as a increases.

Ideally some experimental measurements could give much more insight into these types of flows. Nevertheless, it appears from the experiments demonstrated up to now, that it is really difficult, if not impossible, to conduct accurate measurements at very high spin rates.

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